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--- Summary ---
(3)
[19]

GRAND TOTAL: 75

Copyright reserved ANSWER SHEET

Name and surname: ____

QUESTION 3.2

02468101214
January February March April May JuneNumber of cars
MonthsNumber of cars sold by Mr. Botha from January -June
White cars
Other colour cars

Demo
NW/JUNE/MATLIT/ EMIS/6*******
1

MARKS: 75

Symbol Explanation
M Method
CA Continuous accuracy
A Accuracy
C Conversion
S Simplification
RT/RG/RM Read from Table / Read from graph/ Read from Map
SF Substitution in formula
O Opinion/Example / Deduction/conclusion
P Penalise for example no units / incorrect rounding etc
R Rounding
J Justification/ Motivation/ Supply a Reason

NATIONAL
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GRADE 11
MATHEMATICAL LITERACY P1
JUNE EXAMINATION MEMO
2019

2018

Demo
NW/JUNE/MATLIT/ EMIS/6*******
2
QUESTION 1 [15]
OUES SOLUTIONS EXPLANATIONS L1-4
1.1.1 Rent ■
Cell phone contract ■

(2)
2 A 1
1.1.2 R 250 , 00 × 4 = R 1 000, 00■
R 1 000, 00 + R 600, 00■
R 1 600, 00■

(3)
1M
1M
1A 2
1.1.3 B = R150, 00 + R 500, 00 + R 300, 00+ R 120, 00 ■
= R 1 070,00■

(2) 1M
1A 2
1.1.4
1 070 ,00
1 600 ,00×100■
= 66,875 %■
= 67%■

(3) 1M
1A
1R 3
1.2.1 4:30 – 9:00■
= 5hours 30minutes ■

(2) 1M
1A 2
12.2 5hours ×60minutes = 3 00minutes ■
300minutes + 30minutes ■
= 330minutes ■

(3)
1M
1M
1A 3

QUESTION 2
[19]
2.1.1 Cost = R450 +(R4×number of loaves) ■■■■

2Correct formula
(2)
2
2.1.2 Value for A= R450 + (R4 ×300) ■■
= R1 650,00■
Value for B= R6.50 × R180 ■
=R1 170,00■
1 Substitution
1 Answer
1 Substitution
1 Answer
(4)
2
2.1.3
Cost of 50 loaves = R450 + (R4 × 50) ■■
= R650, 00■ 1 Substitution
1 Answer (2)

Demo

Graph: See attach ed page

1 heading
1 A (0;450)
1 A (0;0)
1 breakeven point
1 labelling both
axis
1 labelling both
graphs

1A joining the

points
(7)
2
2.1.5 Coordinates (180 ■; R1 170) ■■
It means that both cost and income for baking and selling 180 loaves is the same amount which is R1 170.
■ 1A direction and street
1A direction and street
1A destination
(4) L3
4.2.3 Actual distance = 6 ×100 000 cm = 600 000 c m■
Distance on map = 600 000 ■■
75 000■
= 8cm■
OR
Actual distance = 75 000 ■■
100 000■
= 0,75 km
Distance on map = 6 × 1 ■■
0,75■
= 8cm■

1C conversion
1M using scale
1A answer
(3)
L3
TOTAL MARK : 75

Demo
NW/JUNE/MATLIT/ EMIS/6*******
7
PAPER ANALYSIS
QUESTION L2 L3 L4 FINANCE MEASUREMENT MAPS, PLANS AND OTHER REPRESENTATIONS OF THE REAL WORLD
1.1 2 2
1.2 3 3
1.3 8 8
1.4 3 3
1.5 5 5
2.1.1 2 2
2.1.2 3 3
2.1.3 6 6

2.2 8 8
3.1.1   2 2
3.1.2   2 2
3.1.3 4   4
3.2.1 3   3
3.2.2   2 2
4.1.1   2 2
4.1.2   4 4
4.1.3(a) 3   3
4.1.3(b)   4 4
4.2.1 2   2
4.2.2   4   4
4.2.3   3   3
TOTAL
MARK 20 29 26 45 21 9
EXPECTED
MARK 15 30 30
ATUAL % 25 39 35
EXPECTED
% 25 35 40

Demo
NW/JUNE/MATLIT/ EMIS/6*******

MARKS: 75
TIME: 1h 30 minutes

This question paper consist of 8 pages, an annexure and an answer sheet
MATHEMATICAL LITERACY PAPER 1
JUNE EXAM 2019
9
m
MEMORANDUM
NATIONAL
SENIOR CERTIFICATE
GRADE 11
Mathematical literacy/P 1 NSC-Grade 11 NW/June 2019

Demo
NW/JUNE/MATLIT/ EMIS/6*******
2
INSTRUCTIONS AND INFORMATION
1.
ANSWER SHEET 1
QUESTION 1.3
TABLE 1: Phemelo’s Income

Number of passengers 0 5 10 15 20 25
Income

050100150200250300350400450500
0 5 10 15 20 25Amount in rand
Number of passengersPhemelo's Expenses and Income
Expenses

Demo

NAME OF LEARNER………………………………………… CLASS……..

ANSWER SHEET 2

QUESTION 2.2

Complete the table below

TABLE 2: Water tariff for Tshwane Municipality with effect from 1 July 2018

Water usage in kilolitres (k■)	Tariff per kilolitre	Calculations	Cost
0 to 6 k■ per 30 days period	9,54		
7 to 12 k■ per 30 days period	13,62	R13,62× 6	R81,72
13 to 18 k■ per 30 days period	17,89		
19 to 24 k■ per 30 days period	20,70	R20,70× 6	R124,24
25 to 30 k■ per 30 days period	23,66	R23,66× 6	R141,96
31 to 42 k■ per 30 days period	25,57	R25,57× 12	R306,84
43 to 72 k■ per 30 days period	27,36		
More than 72 k■ per 30 days period	29,29		
Total cost excluding VAT			

Total cost including VAT =

Demo

ANNEXURE

QUESTION 4.2

A MAP OF PART OF PRETORIA

Scale 1: 75 000

Demo

MARKS: 75
Symbol Explanation
M Method
CA Continuous accuracy
A Accuracy
C Conversion
S Simplification
RT/RG/RM Read from Table / Read from graph/ Read from Map
SF Substitution in formula
O Opinion/Example / Deduction/conclusion
P Penalise for example no units / incorrect rounding etc
R Rounding
J Justification/ Motivation/ Supply a Reason
F Formula
NPR No penalty for rounding OR ommiting units
AO Answer only

NATIONAL
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GRADE 11
MATHEMATICAL LITERACY PAPER 1
JUNE EXAMINATION 2018
MARKING GUIDELINE

2018

Demo

QUESTION 1 [13]
TL
1.1.1
Mass = 500
1000ü
= 0,5kgü
1M division
1 Answer

(2)
1
1.1.2 ■■■ ■■■■■■ = 2

3 ×250ü
= 166,67 ü
= 167 ml oil needed ü

1Mmultiplying
1Answer
1Rounding

(3) 1
1.1.3
Total muffins = 12×2ü
= 24ü
1Multipying
1Answer

(2) 1
1.2.1 0 to 1 hour ü
Sundays and public holidays ü

2 Answer

(2)
1
1.2.2 Total time = 13: 25 – 08: 45 ü
= 4 hours 40 minutes ü
1Subtraction
1Answer

(2)

1
1.2.3 Amount paid = R6,00 üü

2Anwser

(2) 1

QUESTION 2 [16]
2.1
Total Cost = R 2 400 × R 50, 00 + n, where n is the number of p eople
attending the conference.
üü

2 Answer

(2) 2
2.2 A= R2 400 + 50 × 20 ü
= R2 400 +R 1000
= R3 400 ü
1Substitution
1Answer

(2)
2
2.3 I = 130 × n
10 400 = 130 × n
n = 80 1Substitution
1Answer
(2) 2

Demo

1M heading 1M (0, 2 400)
1M labelling both axis 1M(100,7 400)
1M (0,0) 2 M joining the points
1M(100, 13 000) 1M labelling both graphs
(8)

2.5 30 Tickets üü
2A answer

(2) 2
QUESTION 3 [23]
3.1.1 R1 000, 61üü

2RT

(2) 1
3.12 R2 00,00 üü
2RT

(2) 1
3.1.3 Life style magazine feeüü
2A Answer

(2) 1
3.1.4
■■■■■■■■ ■■■■ =■23,73
■1 000 ,61×100ü
= 2,371553352%
=2,38%ü
1substitution
1A
NPR

(2)
2

3.2 12%
12%=1% Per month ü

■■■■■ ■■■■■=(1
100×R13500) + ,R13500, 00 ü 1Conversion
1Substitution
1A answe r
1A answer

020004000600080001000012000014000
0 20 40 60 80 100 120Amouunt in Rands
Number of PeopleIncome and Cost
Cost
Income

= R13 635,00 ü

■■■■■■ ■■■■■= (1
100×R13 635) + R13 635, 00
= R13 771,35 ü

Third month = (1
100×R13 771 ,35) + R13 771, 35
= R13 909,02 ü

OR

First month = 1,01 × R13 500ü
= R13 635 ü
Second month = 1,01 × R13 635
= R13 771,35ü
Third month = 1,01 × R13 771,35
= R13 909,02 ü ü
1A answer

(5) 3
3.3.1 6kl = 0ü
6kl = R8,04 ×6=R48 ,24ü
6kl = R10,55 ×6=R63 ,30
6kl = R12,21 ×6=R73 ,26
3kl = R13,95 ×3=R41 ,85ü
Total paid = R48,24+ R63,30 + R73,26 + R41,85 ü
= R226,65ü
1A answer
1Method
1A answer
1A answer
1A answer

(5)
2

3.3.2 New charges =
10017 × R44,82 + R44,82 ü
= R52,44 ü
1Method
1Answer

(2)
2
3.4 Earning in other country =
93,1625000R üü
= £1476,67ü
1 numerator
1denominator
1A answer
(3)
1
QUESTION 4
[23]
4.1.1 Radius = 150
2ü

= 75c mü

1Dividing
1A answer
(2)
2
4.1.2 Volume = ■■2h
V = 3,142 × (75■■)2ü× 250ü■■
V = 4418437,5 c ■3 ü

1000c ■3= 1litre

V = 4418437 ,5
1000
V = 4418,4375 ü
V = 4418 ,43litresü
1Conversion
1Substitution
1A answer
1A answer
1Rounding

(5)

3

4.1.3 H = 2,5m
Radius = 75
100
= 0,75mü
Surface area = 2πr(r + h)
= 2 × 3,142 × 0,75(0,75 + 2,5) ■2ü
= 2 × 3,142 × 0,75(3,25) ■2
= 15,31725 ■2 ü
=15,32 ■2ü
1Radius
1Substitution
1A answer
1units

(4)

3

4.2.1 8 Medical stations üü
23 Refreshment stations üü 2A answer
2A answer
(4) 1
4.2.2 44 kilometres üü
2A answer
(2) 1
4.2.3 University of Cape town üü

2A answer
(2) 1
4.2.4
Speed = ■■■■■■■■
■■■■

Speed = 56
3,25ü ü
Speed = 17,23 ü km/hü
 1Substitution
1 Conversion
1A answer
1Unit
(4) 3

MARKS : 75
CODES EXPLANATION
M Method
M/A Method with accuracy
CA Consistent accuracy
A Accuracy
C Conversion
S Simplification
RT/RG/RD Read from table/graph/diagram
AO Answer only
SF Substitution in a formula
O Opinion
P Penalty e.g.
■
OR
1A percentage
1M multiplying with 85%
1CA total cost
1O opinion
(4)
2.3.2
(a) Percentage = 350
10 000×100%■
= 3,5 %■ 1M multiplying and
dividing
1A percentage
(2) F
L2
2.3.2
(b) Total monthly repayments = R764,84 ×36■
= 27 534,24■
Interest = R27 534,24 – R10 000
= R17 534,24 ■ 1MA multiplying the
correct value s
1A total repayment

1CA interest
(3) F
L2
2.3.3 In case of death ■■
OR
Permanent disability ■■
OR
Retrenchment ■■
OR
Diagnosed with critical illness ■■
Accept any relevant reason 2O opinion

(2) F
L4
2.4 ■F = 1,8 × ■C + 32■
140■= 1,8 × ■C + 32■ ■
■C = 108■
1,8■
■C = 60■■ 1SF substitution

1S changing the subject of
the formula
1A temperature
(3) M
L3
QUESTION 3 [17 MARKS]
3.1 Measurement = 5 ,2 cm■ (Accept 5,1 cm to 5,3 cm)
Scale 2 ,1 cm : 300 km ■ (Accept 2 cm to 2, 2 cm)
Actual distance = 5,2 cm ×300 km
2,1 ■■■
= 742, 86 km■
OR

Measurement = 52 mm■ (Accept 51 mm to 53 mm
Scale 2 1 mm : 300 km ■ (Accept 20 mm to 2 2 mm)
Actual distance = 52 mm ×300 km
21 ■■■
= 742,86 km■ 1A measured distance
1A measured scale
1M using scale

1CA actual distance

OR

1A measured distance

1A measured scale
1M using scale
1CA actual distance
NPR (4) MP
L3
3.2 Cost of half full tank = R14,01 ×30■ 1M multiplying
1A cost F
L4

Demo

= R420,30 ■

The gauge was properly working ■

OR

No.

Note: 1 feet = 12 inches
1 metre = 39,37 inches

The rates for riding in a cable car are given as follows:

Adults: 7,6 $
Children below 14 years: 4,8 $
Note: R1 = 0,080 $

Demo

ANNEXURE A

QUESTION 2.3

LEBALLO FINANCIAL ASSISTANCE

Personal loan repayment table

Loan amount	MONTHLY PAYMENT FOR DIFFERENT PERIODS				
	12 months	24 months	36 months	48 months	60 months
R4 000	R936,43	R519,28	R384,42	R315,60	R276,76
R10 000	R1 872,85	R1 038,55	R764,84	R613,21	R553,52
R20 000	R2 809, 28	R1 557,83	R1 147,27	R996,81	R830,27
R30 000	R3 745,70	R2 077,10	R1 529,69	R1 262,42	R1 107,03

The amounts shown above are approximates and will vary according to interest rate fluctuations.

--- Full Combined Text ---

MARKS: 75

TIME: 1½ hours

This question paper consists of 9 pages and an answer sheet.

PROVINCIAL ASSESSMENT
GRADE 11

MATHEMATICAL LITERACY P1
JUNE 2024

INSTRUCTION S AND INFORMATION

1. This question paper consists of FOUR questions. Answer ALL the questions.

2. Answer QUESTION 3.2 on the given ANSWER SHEET . Write your name on the space provided and hand in the ANSWER SHEET with your ANS WER BOOK.

3. Number the answers correctly according to the numbering system used in this question paper.

4. Start EACH question on a NEW page.

5. You may use a n approved calculator (non -programmable and non -graphical), unless stated otherwise.

6. Show ALL calculations clearly.

7. Round off ALL final answers appropriately according to the given context, unless stated otherwise.

8. Indicate units of measurement, where applicable.

9. Diagrams are NOT necessarily drawn to scale, unless stated otherwise.

10. Write neatly and legibly.

QUESTION 1

1.1 Various options are provided as possible answers to the following questions. Choose the answer and write only the letter (A –D) next to the question number s (1.1.1 to 1.1. 4), e.g. 1.1. 5 C.

1.1.1 A variable whose values depend on the values of another variable.

A Independent variable
B Discrete data
C Dependent variable (2)

1.1.2 Expenses that stay that same amount every month .

A Fixed expenses
B Irregular expenses
C Variable exp enses (2)

1.1.3 Conducting research by c arefully watching and recording behaviour or some characteristic .

A Interview
B Observation
C Questionnaire (2)

1.1.4 This type of graphs is useful for comparing the type of relationship that exists between two different variables or quantities for which no obvious pattern is visible.

A Histogram
B Line graph
C Scatter plot graph (2)

1.2

Use the salary slip above to answer the questions that follow.

1.2.1 Calculate Mr. Combrink’s gross income. (2)

1.2.2 UIF stands for ...

A Unemployment Insurance Federation
B Unemployment Insurance Fund
C Unemployment Insurance Fundraising (2)

1.2.3 Name the income paid to the employee after all the deductions have been made. (2)
[14]

Z5Z Computers

EMPLOYEE : Mr. AB Combrink

ID: 890430 1234 567
EMPLOYEE NO.: C159
OCCUPATION: Sales Manager
PAYMENT PERIOD: 1 October – 31 October 2023
PAYMENT SUMMARY
INCOME DEDUCTIONS
Basic salary R14 000,00
Commission R11 500,00
PAYE Tax R2 385,00
UIF R140,00
Pension R1 125,00
Medical aid R1 850,00
GROSS INCOME TOTAL DEDUCTIONS R5 500,00
NET INCOME R20 000,00

QUESTION 2

2.1 Mrs. Joan Botha is looking for a cell phone to purchase. She does some research and gets the following quote for a cell phone from Super Save Cell Phones.

Super Save Cell
Phones

Date: 15 March 2024 FOR : Mrs. Joan Botha
159 Doring Street
Wonderfontein
Thank you for your enquiries 1101

PRODUCT:

Samsung A52 R2 607,83
SUBTOTAL R2 607,83
VAT @ 15% R391,17
TOTAL R2 999,00

Please take note of the following :
- Ordering does not guarantee the specific colour of the phone.
- Price valid for 14 days.

Use the information above to answer the questions that follow.

2.1.1 Name the make of the cell phone for which Mrs. Botha got a quote. (2)

2.1.2 Show how the VAT amount was calculated. (2)

2.1.3 Write out the total amount for the cell phone in words. (2)

2.1.4 a) Until when is this quotation valid? (2)

b) Explain why quotes are only valid for a certain time period. (2)

2.2 Betty and John Botha are sister and brother. Their parents gave each of them a cell phone because they are going to high school. Betty received a prepaid price plan option and John received a monthly contract price plan option. The graph below shows the two price plan options of Betty and John.

TABLE 1: BETTY AND JOHN'S CELL PHONE PRICE PLAN OPTIONS

Number of minutes	0	20	60	80	100	150	160	200
Prepaid price plan option (A) – Betty	R0	R40	R120	R160	R200	R300	R320	R400
Contract price plan option (B) – John	R200	R200	R200	R200	R224	R284	R296	R344

Use the information above to answer the questions that follow.

2.2.1 Name the indep endent variable on the graph . (2)

2.2.2 What are the fixed cost for the contract price plan option? (2)

050100150200250300350400450
0 20 40 60 80 100 120 140 160 180 200Total cost in Rand
Number of minutesBetty and John's cell phone price plan options
B

2.2.3 Use the table and grap h A to write a formula to calculate the total cost of Betty's price plan in the form :

■■■■■ ■■■■ = … (3)

2.2.4 According the graphs, at how many minutes will the total cost for both Betty and John's price plan options be the same? (2)

2.2.5 Calculate the cost per minute for the contract price plan option. (3)
[22]

QUESTION 3

3.1 Mr. Johan Botha sells second -hand cars. He mainly sells white cars and minimally other coloured cars. Below is the calendar showing the colour and number of cars he sold during January to June months.

TABLE 2: NUMBER OF CARS SOLD FROM JANUARY TO JUNE
JANUARY
7 white cars
2 other colour cars FEBRUARY
9 white cars
3 other colour cars MARCH
12 white cars
5 other colour cars
APRIL
8 white cars
2 other colour cars MAY
13 white cars
4 other colour cars JUNE
11 white cars
2 other colour cars

Use the information above to answer the following questions .

3.1.1 How many other colour cars did Mr. Botha sell during the first six months of the year ? (2)

3.1.2 Calculate the average number of white cars per month that he sold from January to June. (3)

3.1.3 Determine the median number of white cars sold in the six months . (3)

3.1.4 In the next month (July) the largest number of other colour cars is sold. Determine this number of other colour cars that is sold during the month of July if the range of the other colour cars is 8 . (3)

3.1.5 Determine the probability of selling a white car in the month of March out of all the white cars sold in the six months. Give your answer as a decimal number . (3)

3.2 On the given ANSWER SHEET , complete the compound bar graph to indicate the number of cars sold by Mr. Botha during the first six months of the year . (4)

3.3 Except for the colour of a car, name ONE other factor that a person will look at when deciding to buy a car . (2)

[20]

QUESTION 4

The Botha family keeps a record of their water consumption for the month of June. Below is the number of kilolitres water used during the month.

TABLE 3: NUMBER OF KILOLITRES USED

WATER USAGE	KILOLITRES
Dish washing and laundry	5,5
Cleaning of the house	2,5

Shower and bath	4,8
Toilet	1,3
Outside water usage	5,9

The table below shows the water tariffs charged by die municipality in the town where the Botha family lives.

TABLE 4: WATER TARIFFS

BLOCKS	CENTS / k■ EXCLUDING 15% VAT
Block 1: 0 – 6 k■	0,00c
Block 2: 7 – 15 k■	1 095c
Block 3: 16 – 25 k■	1 248c
Block 4: 25 – 50 k■	1 645c
Block 5: 51 k■ and more	2 103c

Use the information above to answer the following questions.

4.1 Calculate the percentage of the total water consumption for the month spent on cleaning the house? (3)

4.2 If 2,8 kilolitres of water are used to bath and the rest to shower, determine the probability to use bath water during the month of June. Give your final answer as ‘n simplified fraction . (3)

4.3 Mr. Botha claims that the month’s total water bill (VAT included) for the family will be more than R200.

Verify showing all calculations, whether Mr. Botha’s claim is VALID . (7)

4.4
Is the data displayed in TABLE 3 discreet or continuous? Motivate your answer. (3)

4.5 Betty wants to display the data in TABLE 3 in a graph. Give her advice on the type of graph to use and whether she must use the actual values or percentages to draw the graph. (3)
[19]

GRAND TOTAL: 75

ANSWER SHEET

Name and surname: ____

QUESTION 3.2

02468101214
January February March April May JuneNumber of cars
MonthsNumber of cars sold by Mr. Botha from January -June
White cars
Other colour cars

MARKS: 75

Symbol Explanation
M Method
CA Continuous accuracy
A Accuracy
C Conversion
S Simplification
RT/RG/RM Read from Table / Read from graph/ Read from Map
SF Substitution in formula
O Opinion/Example / Deduction/conclusion
P Penalise for example no units / incorrect rounding etc
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2019

2018

QUESTION 1 [15]

OUES SOLUTIONS EXPLANATIONS L1-4
1.1.1 Rent ■
Cell phone contract ■

(2)

2 A 1
1.1.2 R 250 , 00 × 4 = R 1 000, 00■
R 1 000, 00 + R 600, 00■
R 1 600, 00■

(3)
1M
1M
1A 2
1.1.3 B = R150, 00 + R 500, 00 + R 300, 00+ R 120, 00 ■
= R 1 070,00■

(2) 1M
1A 2
1.1.4
1 070 ,00
1 600 ,00×100■
= 66,875 %■
= 67%■

(3) 1M
1A
1R 3
1.2.1 4:30 – 9:00■
= 5hours 30minutes ■

(2) 1M
1A 2
12.2 5hours ×60minutes = 3 00minutes ■
300minutes + 30minutes ■
= 330minutes ■

(3)
1M
1M
1A 3

QUESTION 2
[19]
2.1.1 Cost = R450 +(R4×number of loaves) ■■■■

2Correct formula
(2)
2
2.1.2 Value for A= R450 + (R4 ×300) ■■
= R1 650,00■
Value for B= R6.50 × R180 ■
=R1 170,00■
1 Substitution
1 Answer
1 Substitution
1 Answer
(4)
2
2.1.3
Cost of 50 loaves = R450 + (R4 × 50) ■■
= R650, 00■ 1 Substitution
1 Answer (2)

3

Graph: See attach ed page

1 heading
1 A (0;450)
1 A (0;0)
1 breakeven point
1 labelling both
axis
1 labelling both
graphs

1A joining the
points
(7)
2
2.1.5 Coordinates (180 ■; R1 170) ■■
It means that both cost and income for baking and selling
180 loaves is the same amount which is R1 170. ■■■■
2 coordinates
2 Meaning

(4)
2

QUESTION 3
[20]
3.1.1 Two items ■■

(2) 2A 1
3.12 R 499, 00 ×33
100 ■
R 164, 00■

(2) 1M
1M
2
3.1.3 Total including VAT
R 768, 00 ×15
100■
= R 115, 20 + R 768, 00■
= R 883, 40 ■

(3) 1Mmultiplication
1Maddition
1Aanswer

2
3.2.1 Cost of water used
6kl = R0, 00
24kl × 6,48■ = R155, 52 ■
5kl × 16,20 = R81,00 ■
R155, 52 +R81,70■ =R236,52
R236,52 + R80,70■ + R17,15■
= R324,37 ■
(7) 1Multiplication
1Aanswer
1Aanswer
1Maddition

1Madding R80,70
1Madding R17,15
1Aanswer
2

3.2.2 New price = R80,70 ×15
100 =R 12,01■
= R80,82 + R12, 01■
= R 92, 81■
(3) 1Mmultiplication
1Maddition
1Aanswer
2

3.3
1■■ ■■■■ = (7,5
100×R120 000 ,00)+ R 120 00,00 ■

= R129 000 , 00■

2■■ ■■■■ =   7,5
100×R129 000 ,00 + R129 000, 00
= R138 675 ,00 ■

3■■ ■■■■ =   7,5
100×R139 675 ,00 + R139 675,00
= R149 075,63 ■ (4)
1Multiplication
1Aanswer
1Aanswer
1Aanswer

3

QUESTION 4
[15]
4.1.1   Radius = 14
2 ■
= 7cm■ (2)
1dividing by 2
1Aanswer
1
4.1.2
■■■■■■ = ■■2× h
V= 3,142 × (7■■)2+24 ■■2■
V = 3694,99 c ■3 ■
V= 3694.99 c ■3 - 3456 c ■3 ■
V= 238.99 ■3   ■
(4) 1Substitution
1Aanswer
1M
1A

3

4.1.3 Volume = (■■■■)2 × height
= 3456 c ■3 = (■)2× 24cm■
(■)2 = 3456 c ■3 ÷ 24cm■
(■)2= 144 ■■3
(■)2= √ 144■
A = 12 ■

(4) 1S
1M
1M
1A

3

4.2.1 Boipelo nature reserve ■■ (2) 2A 1
4.2.2 N 12■, N 14 ■and N 4■ (3) 3A 1
4.2.3 Turn into R505 ■ and continue until you reach Lichtenburg ■. Turn into R52 ■ and join N14 ■ and continue until you reach Sannieshof. ■ ■
(6) 2

05001000150020002500
0 60 120 180 240 300Cost in rands
Number of loaves of breadCost and Income of loaves of Bread
Costs

SYMBOL EXPLANATION
M Method
M/A Method with accuracy
CA Consistent accuracy
A Accuracy
C Conversion
S Simplification
RT/RG Reading from a table/Reading from a graph
F Choosing the correct formula
SF Correct substitution in a formula

O Opinion/Example
P Penalty, e.g. for no units, incorrect rounding off etc.
R Rounding off
J Justification/Reason

MARKS: 75
This marking guidelines consists of 7 pages .

NATIONAL
SENIOR CERTIFICATE
MATHEMATICAL LITERACY P2

MARKING GUIDELINES
JUNE 2019

MARKING GUIDELINES
Ques Solution Explan ation Taxonomy
levels
QUESTION 1 [21 MARKS]
1.1 Expense (R) = R200 ■ + R5 ×number of passengers ■ 1A fixed value
1MA multiplication
(2) L2
1.2 Expense (R) = R200 + R5 ×7■
= R200 + R35 ■
= R235■ 1SF substitution
1S simplification
1A answer
(3) L2
1.3 TABLE: Phemelo's Income

Number of passengers	0	5	10	15	20	25
■■■ Income	0	100	200	300	400	500

Table :
1A any two correct
points
1A any two correct
points
1A any two correct
points

Graph :
1CA starting point
1CA break -even pint
1CA end point
1A joining the points
1A naming graph
(8) L3
1.4 No, ■ because the break -even point is between 13 and 15 passengers.
Accept the break -even poin t is between 12,5 and 15
passengers.
1O opinion
2J justification
(3) L4

1.5 ■F = 1,8 ×37■ + 32■■
= 66,6■■+ 32■
= 98,6■■
= 99■■
The passenger is correct . ■
1SF substitution
1S simplification
1A answer
1R rounding
1O opinion
L4 050100150200250300350400450500
0 5 10 15 20 25Amount in rand
Number of passengersPhemelo's Expenses and Income
ExpensesIncome
■■■
■
■

OR
99■ ■ = 1,8 ×■C + 32■
1,8×■C = 99■- 32■
■C = 67■÷1,8■
= 37,2222 ■■
= 37■■
The passenger is correct. ■
OR
1SF substitution

1S simplification

1A answer
1R rounding
1O opinion
(5)
QUESTION 2 [19 MARKS]
2.1.1 To minimise water that spill out of the pool when people are swimming. ■■
OR
When people get into the swimming pool the water will rise to a higher level.
Accept any relevant reason. 2A answer

(2) L4
2.1.2 Area = side × side
= 32,8084 ×32,8084■
= 1 076,39111 ■
≈ 1 077 feet2■■
1SF substitution

1A answer

1R rounding (3) L2
2.1.3 radius = 32,8084 ÷2 = 16,4042 feet ■
radius = 16,4042 ÷3,28084 ■=5 m■
Volume of circular = π × r2 × h
= 3,142 × (5)2 × 2 m■
= 157,1 m3 ■
= 157,1 kilolitres ■
OR

radius = 32,8084 ÷2 = 16,4042 feet ■
height = 2 × 3,28084 ■=6,56168 m ■
Volume of circular = π × r2 × h
= 3,142 × (16,4042)2 × 6,56168 m ■
= 5547,934675 feet3 ■
=
■■33
3,280845m 5547,93467
= 157,1 kilolitres ■
1A radius in feet
1C conversion
1A radius in metres

1SF substitution
1S simplification
1A volume kilolitres
OR
1A height in feet
1C conversion
1A height in metres

1SF substitution
1S simplification

1A volume in kilolitres
(6) L3

2.2 Calc. Cost
R9,54× 6 ■ R57,24■
R13,62× 6 R81,72
R17,89× 6 R107,34■
R20,70× 6 R124,24
R23,66× 6 R141,96
R25,57× 12 R306,84
R27,36× 30 R820,80■
R29,29× 78 R2 284,62
Total cost R3 924,76 ■
Total cost including VAT = R3 924,76 ×115
100■
= R4 513,47 ■
OR
Total cost including VAT
= R3 924,76 ×15
100 + R3 924,76
= R588,71 ■ + R3 924,76
= R4 513,47 ■
1M multiplication
1CA answer

1A calculation and cost
1A calculation and cost
1A calculation and cost
1CA total cost

1M calculating VAT
1CA cost including
VAT

OR
1CA VAT amount
1CA cost including
VAT
(8) L3

QUESTION 3 [13 MARKS]
3.1.1 The school did not receive the expected number of learners ■■
OR
Some learners were transferred to other schools before the end of the year. ■■
OR
Less number of learners enrolled ■■ 2Answer

(2) L4
3.1.2 Water ■and sanitation■
Telephones ■
Electricity ■
Gardening ■
Cleaning■
Accept any relevant answer 2A any two correct
answer

(2) L4
3.1.3 Percentage = 420 000
993 500 ■×100%■
= 42%■
The claim is valid. ■
OR
Amount = 40
100× R993 500 ■■
= R397 400 ■ (it is less than R420 000)
The claim is valid ■ 1RT correct values
1M multiplication
1A answer
1O opinion
OR
1RT correct value

1M multiplication
1A answer
1O opinion (4) L4

3.2.1 Time = 30 min + 45 minutes + 45 Minutes ■
= 120 minutes ■
= 2 hours ■ 1M adding
1A time in minutes
1A time in hours
(3) L2

3.2.2 To prepare children to sleep. ■■
OR
To develop listening skill ■■
OR
To improve vocabulary ■■
OR
To develop oral communication skills ■■
OR
Helps kids to bond with their teachers. ■■
Accept any relevant reason 2J reasoning

(2) L4

QUESTION 4 [22 MARKS]

4.1.1 The decrease in the general price of goods and services ■■
OR
It means deflation ■■
OR
Inflation decline 2A answer

(2) L4

4.1.2 Amount = 75
14,2417■
= 5,26622524€ ■
= 5,26622524 ×1,15$■
= 6,056159026 $ ■
1M dividing
1S simplification
1M multiplication
1A answer
NPR
(4) L3

4.1.3
(a) 2016 monthly rental fee = 100
110 ,61■× 1654 EGP ■
= 1 495,344001 EGP ■

OR
2016 monthly rental fee increase = 10,61
110 ,61× 1654 EGP ■
= 158,6559986 ■
2016 monthly rental fee = 1 654 – 158,6559986
= 1 495,344001EGP ■ 1A fraction

1M multiplication
1A answer
NPR
OR
1M multiplication
1S simplification
1A answer
NPR (3) L2
4.1.3
(b) South Africa:
% increase = ■■■■■■ (2018)–■■■■■■ (2016)
■■■■■■ ■■ 2016 ×100%
% increase = ■7 175 – ■6 350
■6 350 ×100%■
= 13%■
Egypt:
% increase = ■■■■■■ (2018)–■■■■■■ (2016)
■■■■■■ ■■ 2016 ×100%
% increase = 1654 –1495 ,3
1495 ,3 ×100%
= 10,6%■
The claim is not valid. ■

1SF substitution

1A answer

1CA answer
1Opinion
NPR
(4) L4

4.2.1 Up■■ OR above ■■Church Square 2A answer
(2) L2
4.2.2 Take Rissik street, ■ turn right into Nelson Mandela drive, ■ turn left into Pretorius Street, ■destination is on the right . ■
OR
Go north towards Park street ■, turn left into Schoeman street immediately right into Nelson Mandela, ■ turn left into Pretorius street, ■ destination in on the right. ■ 1A direction and street
1A direction and street
1A destination
(4) L3
4.2.3 Actual distance = 6 ×100 000 cm = 600 000 c m■
Distance on map = 600 000 ■■
75 000■
= 8cm■
OR
Actual distance = 75 000 ■■
100 000■
= 0,75 km

Distance on map = 6 × 1 ■■
0,75■
= 8cm■

1C conversion
1M using scale
1A answer
(3)
L3
TOTAL MARK : 75

MATHEMATICAL LITERACY/P2 NSC NW/JUNE 2019
GRADE 11

Demo
NW/JUNE/MATLIT/ EMIS/6*******
7
PAPER ANALYSIS
QUESTION L2 L3 L4 FINANCE MEASUREMENT MAPS, PLANS AND
OTHER
REPRESENTATIONS
OF THE REAL WORLD
1.1 2 2
1.2 3 3
1.3 8 8
1.4 3 3
1.5 5 5
2.1.1 2 2
2.1.2 3 3
2.1.3 6 6
2.2 8 8
3.1.1 2 2
3.1.2 2 2
3.1.3 4 4
3.2.1 3 3
3.2.2 2 2
4.1.1 2 2
4.1.2 4 4
4.1.3(a) 3 3
4.1.3(b) 4 4
4.2.1 2 2
4.2.2 4 4
4.2.3 3 3
TOTAL
MARK 20 29 26 45 21 9
EXPECTED
MARK 15 30 30
ATUAL % 25 39 35
EXPECTED
% 25 35 40

MARKS: 75
TIME: 1h 30 minutes

This question paper consist of 8 pages, an annexure and an answer sheet

MATHEMATICAL LITERACY PAPER 1
JUNE EXAM 2019
9
m
MEMORANDUM
NATIONAL
SENIOR CERTIFICATE
GRADE 11

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FOUR questions. Answer ALL the questions.
2. Use ANNEXURE to answer QUESTION 4.2 and the ANSWER SHEET to answer QUESTION 2.1.3
3. Number the answers correctly according to the numbering system used in the question paper.
4. ALL the calculations must be clearly shown.
5. An approved calculator (non - programmable and non - graphical) may be used, unless stated otherwise.
6. Round off ALL final answers appropriately according to the given context, unless stated otherwise.
7. Indicate units of measurement , where applicable.
8. Maps and diagrams are NOT necessarily drawn to scale , unless stated otherwise.
9. Write neatly and legibly.

QUESTION 1

1.1

1.1.1 Write down her fixed expenses
(2)

1.1.2 Calculate her total income for the month if February has 4 weeks.
(3)

1.1.3 Calculate the total expenses for this month.
(2)

1.1.4 Express the total expenses as a percentage of the total income . Round your answer to the nearest percentage
(3)

1.2 Mapule plans to visit her cousin who stays in Johannesburg using a bus. The bus will leave her home town at 09: 00 and will arrive at 14: 30.

1.2.1 Calculate the total time that the bus takes to arrive in Johannesburg.
(2)

1.2.2 Convert the answer in 1.2.1 to minutes.
(3)

[15]

Mapule works in a local restaurant in her home town. She earns R 250,00 per week plus tips from generous customers. Below is her income and expenditure statement for the month of February.

INCOME EXPENDITURE
R 250.00 Per week

Tips for one month: R 600. 00

Transport : R 150. 00
Groceries : R 500. 00
Rent : R 300. 00
Airtime : R 120.00

Total Expenses ……….

QUESTION 2
2.1

A local confectionary is spending R450, 00 on electricity and water weekly. The cost of baking one loaf of bread is R4, 00 which includes labour and ingredients. A loaf of bread is sold at R6,50.

The TABLE below shows the weekly cost of baking loaves of bread.

TABLE A: Weekly COST of BAKING LOAVES OF BREAD

Number of loaves	0	60	120	180	240	300
Total cost (in Rand)	R450.00	R690.00	R930.00	R1170.00	R1410.00	A

TABLE B: Weekly INCOME from SELLINGLOAVES OF BREAD

Number of loaves	0	60	120	180	240	300
Total Income (in Rand)	0	R390.00	R780.00	B	R1560.00	R1950.00

2.1.1 Write the formula to calculate the cost of baking loaves.
(2)
2.1.2 Determine the value of A and B. (4)
2.1.3 Use the formula in QUESTION 2.1.1 to calculate the cost for baking 50 loaves. (2)
2.1.4 On the same set of axes, use the values from TABLE A and B, and the ANSWER SHEET provided, to draw the graphs illustrating both INCOME and COST of baking loaves of bread. (7)
2.1.5 Write the coordinates of the break -even point and in the context of the scenario exp lain what it means. (4)

[19]

QUESTION 3
3.1

3.1.1 How many items did Thato buy ?
(2)
3.1.2 Calculate the total discount paid on the items that she bought
(2)
3.1.3 Calculate the total including VAT paid on this transaction
(3)
3.2 Thato is a resident in the Phakisa municipality and below is a tariff on a sliding scale that the municipality uses to charge her for water usage.

- Fixed charge if > 6 kl = R80,70
- Free for infrastructure if > = R7,15

3.2.1. Calculate the cost if Thato uses 35 kl of water charge
(7)
3.2.2 Calculate the new fixed charge if it is increased by 15%
(3)

Below is a till slip that Thato got from a boutique after buying some clothes for herself.

28/11/18 14:22:04 Store No : 362
Transaction. No 108 0005 Employee No : 2913457

Tax invoice no: 1122346
Customer name : Thato Martins

		Rands
19441209 Stipe	Self belted dress	550.00
46458553 Stripe	Twofer Top	499.00
Stripe Top Discount @ 33%		A
Subtotal excl. VAT		768.00
TOTAL PAID		………

Water Usage	Rate per kilolitre (VAT of 15%) inclusive
Up to 6 kl	0
7 kl – 30 kl	R6,48
30.1 kl – 60 kl	R16,20
More than 60 kl	R21,60

3.3 Mrs Tsheko is fixing the roof of her house and has decided to take a personal loan of R 120 000 from the bank. The bank will charge her 7.5% simple interest per annum . Calculate the total that she will repay the bank after 3 years.
(4)

[21]

QUESTION 4

A couple has decided to give tokens of appreciation to all the people who will attend their wedding. They will use cylindrical containers and square containers as shown below to package the tokens . The diameter of the cylindrical container is 14 cm.

Volume = 3456 cm3

4.1.1 Write down the radius of the cylindrical container.
(2)

4.1.2 Show with calculations that the volume of the cylindrical container is 238,99cm more than the volume of the container with the square base.

You may use the formula: Volume = ■■■× h where ■ = 3,142
(4)

4.1.3 Determine A the length in cm of ONE side of the square base. You may use the Formula: Volume of the box = (side) × height
(4)

4.2 Use the map on annexure A to answer the questions which follow.

4.2.1 What is the name of a game reserve situated in Taung
(2)

4.2.2 Write the names of the national roads found on this map
(3)

4.2.3 Modise stays in Zeerust and wants to visit his mother in Sannie shof. Give the shortest route that Modise could use to travel from Zeerust to Sannieshof.
(5)

[21]
Token
s d
Token
s Height
240mm
A Height
24cm

Demo

QUESTION 4 .2
ANNEXTURE

Demo

ANSWER SHEET
QUESTION 2.1.4.
Name:...

020040060080010001200140016001800
0 60 120 180 240 300______________________________
____

Demo

MARKS: 75
TIME: 1,5 hour

This question paper consists of 12 pages including 1 annexure and 2 answer sheet.

NATIONAL
SENIOR CERTIFICATE
MATHEMATICAL LITERACY P2

JUNE 2019

GRADE 11 NATIONAL

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FOUR questions. Answer ALL the questions.

2. 2.1 Use the ANNEXURE in the QUESTION PAPER to answer QUESTION 4.2.

2.2 Answer the following questions on the ANSWER SHEETS attached.
■ QUESTION 1. 3 on the ANSWER SHEET 1
■ QUESTION 2.2 on the ANSWER SHEET 2

3. Number the answers correctly according to the numbering system used in the question paper.

4. An approved calculator (non - programmable and non - graphical) may be used, unless stated otherwise.

5. ALL the calculations must be clearly shown.

6. Round off ALL final answers appropriately according to the given context, unless stated otherwise.

7. Units of measurement MUST be indicated, where applicable.

8. Diagrams are NOT necessarily drawn to scale, unless stated otherwise.

9. Write neatly and legibly.

QUESTION 1

Use the information above to answer the questions that follow.

1.1 Write down the formula that Phemelo can use to calculate his expenses in the form:
Expense (R) = (2)

1.2 Use the formula in Question 1.1 to determine the expense for transporting seven passengers. (3)

1.3 The formula for calculating Phemelo’s income is as follows:
Income = R2 0 × number of passengers

Use the above formula to complete TABLE 1 on the ANSWER SHEET provided.
Hence, draw the graph illustrating Phemelo’s income on the same set of axes with
the graph of his expenses. (8)

1.4 Phemelo claims that for a single trip he only makes profit if he transport a
minimum of 12 passengers. Do you agree with Phemelo? Use the graph to
justify your ans wer. (3)

1.5 On a particular day, the temperature on the dashboard of the taxi was 99■F.
One of the passengers stated that this temperature is 37■C. Verify, using
calculations, whether the passenger is correct.
Note: The temperature is displayed on the dashboard as a whole number .
You may use the formula:
■F = ■,■×■C + 32■ (5)
[21]

Phemelo owns a taxi with a maximum capacity of 25 passengers. His cost per trip include R5
per passenger payable to the taxi association and R200 for fuel. He charges R20 per passenger
who boards his taxi.

QUESTION 2
2.1

Use the information above to answer the questions that follow.

2.1.1 Malebogo states that the pool should be filled to 80% of its capacity.
Give ONE possible reason for her statement. (2)

2.1.2 Calculate the amount of material (to the nearest feet2) needed to cover the entire circular swimming pool .
(3)

2.1.3 Calculate the volume (in kilolitres) of the circular swimming pool.
(6)

Malebogo has a circular swimming pool with a diameter of 32 ,8084 feet and the depth of
2 metres in her yard. She cleans it every month and it costs her a lot of money to refill.
She intends to buy a square covering material for swimming pool so that she no longer

cleans it monthly.
Below it the picture of the swimming pool and the diagram illustrating a swimming pool with the covering material.

You may use the following formulas:
Area of a square = side × side
Volume of a circular prism = $\pi r2h$, where $\pi = 3{,}142$
NOTE : 1 m = 3,28084 feet and 1 m3 = 1 kilolitre
[source: poolwarehouse.com]

Swimming pool
Covering material Diagram Picture

2.2

Use information above to complete TABLE 2 on ANSWER SHEET 2 . (8)
[19]

Assume that Malebogo used 150 kilolitres of water I July 2018.
The table below shows the water tariff for Tshwane Municipality with effect from 1 July 2018.

TABLE 2: Water tariff for Tshwane Municipality with effect from 1 July 2018

Water usage in kilolitres (k■)	Tariff per kilolitre
0 to 6 k■ per 30 days period	9,54
7 to 12 k■ per 30 days period	13,62
13 to 18 k■ per 30 days period	17,89
19 to 24 k■ per 30 days period	20,70
25 to 30 k■ per 30 days period	23,66
31 to 42 k■ per 30 days period	25,57
43 to 72 k■ per 30 days period	27,36
More than 72 k■ per 30 days period	29,29

NOTE: The above tariffs EXCLUDE 15% VAT.

QUESTION 3

3.1

Use TABLE 3 above to answer the questions that follow.

3.1.1 Give ONE possible reason why the actual school fees income in 2017 is not the same as the amount budgeted for in 2017. (2)

3.1.2 Name TWO items that can be classified as services at a pre -school. (2)

3.1.3 One of the SGB members of Ramatla Pre -school claimed that the actual expenses on salaries in 2017 used more than 40% of the total income.
Verify, using calculations, whether the claim is valid. (4)

Below is the Ramatla Pre -School budget for 2017 and 2018 financial years.

TABLE 3 : Ramatla Pre -School Budget

INCOME	2018 budget	2017 actual	2017 budget
School fees	R 831 600	R 693 000	R 739 200
Registration fees	R 25 200	R 21 000	R 22 400
Subsidy	R 275 000	R 275 000	R 275 000
Donations	R 7 600	R 4 500	R 4 800
TOTAL INCOME	R1 139 400	R 993 500	R1 041 400
EXPENSES			
Salaries	R 441 000	R 420 000	R 420 000
Stationery	R 17 000	R 13 750	R 15 000
Services	R 13 200	R 11 472	R 12 000
Transport	R 19 750	R 16 720	R 25 000
Food	R 608 400	R 561 600	R 570 000
TOTAL EXPENSES	R1 099 3 50	R1 023 542	R1 042 000

3.2

Use TABLE 4 above to answer the questions that follow.

3.2.1 Calculate the time (in hours) spent on teaching before lunch on Wednesday. (3)

3.2.2 Give ONE possible reason why there is story telling in the timetable. (2)

[13]

Below is the timetable for Ramatla Pre-school.

TABLE 4 : TIMETABLE FOR RAMATLA PRE-SCHOOL

DAY PERIOD
1 2 3 4 5 6 7
Duration 08:00
to
08:30 08:45
to
09:30 09:45
to
10:30 10:30
to
11:30 11:30
to
12:00 12:00
to
13:00 13:00
to
16:00
Monday Bible
study Drawing Counting
LUNCH Story
telling Sleeping Playing
Tuesday Bible
study Games Language Story
telling Sleeping Playing
Wednesday Bible
study Counting Language Story
telling Sleeping Playing
Thursday Bible
study Games Counting Story
telling Sleeping Playing
Friday Bible
study Language Drawing Story
telling Sleeping Playing

QUESTION 4
4.1

Use TABLE 5 above to answer the questions that follow.

4.1.1 Write down the mean ing of the negative sign for some of the inflation rates. (2)

4.1.2 A South African tourist visited Australia in 2016 and spent R75 on a standard cup of coffee. Determine the amount spent on coffee in dollars. (4)

4.1.3 The annual inflation rate for Egypt remained unchanged for the past two years since 2016.

(a) Calculate the monthly flat rental in 2017 for a flat that was costing 1654 EGP (Egyptian pound) in 2018 . (3)

(b) An Egyptian tourist , Abigo, visited South Africa and stayed in a flat similar to the one in Question 4.1.3(a). During 2016 the monthly rental for this flat was R6 350 and in 2018 was R7 175.

Abigo claims that the percentage monthly rental increase in South Africa is half the percentage monthly rental increase i n Egypt over the same two years.

Verify, using calculations, whether his claim is valid. (4)

You may use the formula:

% increase = ■■■■■■■■ ■■■■■■■ (■■■■)–■■■■■■■■ ■■■■■■■ (■■■■)
■■■■■■■■ ■■■■■■■ ■■ ■■■■ ×100% TABLE 5 below shows a list of the annual inflation rates for some countries during 2016.

TABLE 5 : Annual inflation rates for some countries in 2016

Country	Annual inflation rates (%)
Australia	3,00
Cameroon	1,06
Cyprus	-0,58
Egypt	10,61
Montenegro	-1,20
South Africa	6,3

NOTE : 1 euro (€) = R14,2417 and 1 euro (€) = 1,15 dollar ($)

4.2

Use ANNEXURE provided to answer the questions that follow.

4.2.1 Write down the position of Daspoort on the map relative to Church Square? (2)

4.2.2 Abigo wants to drive from his flat in Sunnyside to the store in Pretoria Central for shopping.
Describe the shortest route he could use. (4)

4.2.3 The actual distance from Abigo' s flat to the store is 6 km. Calculate the distance (in cm) on the map. (3)

[22]

TOTAL MARK : 75

Abigo stayed in Pretoria during his visit in South Africa. He was residing in Sunnyside.

NAME OF LEARNER………………………………………… CLASS………

ANSWER SHEET 1

QUESTION 1.3

TABLE 1: Phemelo's Income

Number of passengers 0 5 10 15 20 25
Income

050100150200250300350400450500
0 5 10 15 20 25Amount in rand
Number of passengersPhemelo's Expenses and Income
Expenses

NAME OF LEARNER………………………………………… CLASS……..

ANSWER SHEET 2

QUESTION 2.2

Complete the table below

TABLE 2: Water tariff for Tshwane Municipality with effect from 1 July 2018

Water usage in kilolitres (k■) Tariff per
kilolitre Calculations Cost

0 to 6 k■ per 30 days period	9,54		
7 to 12 k■ per 30 days period	13,62	R13,62× 6	R81,72
13 to 18 k■ per 30 days period	17,89		
19 to 24 k■ per 30 days period	20,70	R20,70× 6	R124,24
25 to 30 k■ per 30 days period	23,66	R23,66× 6	R141,96
31 to 42 k■ per 30 days period	25,57	R25,57× 12	R306,84
43 to 72 k■ per 30 days period	27,36		
More than 72 k■ per 30 days period	29,29		

Total cost excluding VAT
Total cost including VAT =

ANNEXURE

QUESTION 4.2

A MAP OF PART OF PRETORIA

Scale 1: 75 000

MARKS: 75

Symbol	Explanation
M	Method
CA	Continuous accuracy
A	Accuracy
C	Conversion
S	Simplification

RT/RG/RM Read from Table / Read from graph/ Read from Map
SF Substitution in formula
O Opinion/Example / Deduction/conclusion
P Penalise for example no units / incorrect rounding etc
R Rounding
J Justification/ Motivation/ Supply a Reason
F Formula
NPR No penalty for rounding OR ommiting units
AO Answer only

NATIONAL
SENIOR CERTIFICATE
GRADE 11
MATHEMATICAL LITERACY PAPER 1
JUNE EXAMINATION 2018
MARKING GUIDELINE

2018

QUESTION 1   [13]

TL
1.1.1
Mass = 500
1000ü
=   0,5kgü
1M division
1 Answer

(2)
1
1.1.2   ■■■ ■■■■■■ = 2
3 ×250ü
= 166,67 ü
= 167 ml oil needed ü

1Mmultiplying
1Answer
1Rounding

(3) 1
1.1.3
Total muffins = 12×2ü
= 24ü
1Multipying
1Answer

(2) 1
1.2.1 0 to 1 hour ü
Sundays and public holidays ü

2 Answer

(2)
1
1.2.2 Total time = 13: 25 – 08: 45 ü
= 4 hours 40 minutes ü
1Subtraction
1Answer

(2)

1
1.2.3 Amount paid = R6,00 üü

2Anwser

(2) 1

QUESTION 2 [16]
2.1
Total Cost = R 2 400 × R 50, 00 + n, where n is the number of p eople attending the conference. üü

2 Answer

(2) 2
2.2 A= R2 400 + 50 × 20 ü
= R2 400 +R 1000
= R3 400 ü
1Substitution
1Answer

(2)
2
2.3 I = 130 × n
10 400 = 130 × n
n = 80 1Substitution
1Answer
(2) 2

Demo

1M heading 1M (0, 2 400)
1M labelling both axis 1M(100,7 400)
1M (0,0) 2 M joining the points
1M(100, 13 000) 1M labelling both graphs
(8)

2.5 30 Tickets üü
2A answer

(2) 2

QUESTION 3 [23]
3.1.1 R1 000, 61üü

2RT

(2) 1
3.12 R2 00,00 üü
2RT

(2) 1
3.1.3 Life style magazine feeüü
2A Answer

(2) 1
3.1.4
■■■■■■■■ ■■■■ =■23,73
■1 000 ,61×100ü
= 2,371553352%
=2,38%ü
1substitution
1A
NPR

(2)
2

3.2 12%
12%=1% Per month ü

■■■■■ ■■■■■=(1
100×R13500) + ,R13500, 00 ü 1Conversion
1Substitution
1A answe r
1A answer

020004000600080001000012000140000
0 20 40 60 80 100 120Amouunt in Rands
Number of PeopleIncome and Cost
Cost
Income

Demo

= R13 635,00 ü

■■■■■■ ■■■■■= (1
100×R13 635) + R13 635, 00
= R13 771,35 ü

Third month = (1
100×R13 771 ,35) + R13 771, 35
= R13 909,02 ü

OR
First month = 1,01 × R13 500ü
= R13 635 ü
Second month = 1,01 × R13 635
= R13 771,35ü
Third month = 1,01 × R13 771,35
= R13 909,02 ü ü

1A answer

(5) 3
3.3.1 6kl = 0ü
6kl = R8,04 ×6=R48 ,24ü
6kl = R10,55 ×6=R63 ,30
6kl = R12,21 ×6=R73 ,26
3kl = R13,95 ×3=R41 ,85ü
Total paid = R48,24+ R63,30 + R73,26 + R41,85 ü
= R226,65ü
1A answer
1Method
1A answer
1A answer
1A answer

(5)
2

3.3.2 New charges =
10017 × R44,82 + R44,82 ü
= R52,44 ü
1Method
1Answer

(2)
2
3.4 Earning in other country =
93,1625000R üü
= £1476,67ü
1 numerator
1denominator
1A answer
(3)
1
QUESTION 4
[23]
4.1.1 Radius = 150
2ü
= 75c mü

1Dividing
1A answer
(2)
2
4.1.2 Volume = ■■2h
V = 3,142 × (75■■)2ü× 250ü■■
V = 4418437,5 c ■3 ü

1000c ■3= 1litre

V = 4418437 ,5
1000
V = 4418,4375 ü
V = 4418 ,43litresü
1Conversion
1Substitution
1A answer
1A answer
1Rounding

(5)

3

4.1.3 H = 2,5m
Radius = 75
100
= 0,75mü
Surface area = $2\pi r(r + h)$
= 2 × 3,142 × 0,75(0,75 + 2,5) ■2ü
= 2 × 3,142 × 0,75(3,25) ■2
= 15,31725 ■2 ü
=15,32 ■2ü
1Radius
1Substitution
1A answer
1units

(4)

3

4.2.1 8 Medical stations üü
23 Refreshment stations üü 2A answer
2A answer
(4) 1
4.2.2 44 kilometres üü
2A answer
(2) 1
4.2.3 University of Cape town üü
2A answer
(2) 1
4.2.4
Speed = ■■■■■■■
■■■■

Speed = 56
3,25ü ü
Speed = 17,23 ü km/hü
1Substitution
1 Conversion
1A answer
1Unit
(4) 3

MARKS : 75

CODES EXPLANATION
M Method
M/A Method with accuracy
CA Consistent accuracy
A Accuracy
C Conversion
S Simplification
RT/RG/RD Read from table/graph/diagram
AO Answer only
SF Substitution in a formula
O Opinion
P Penalty e.g. no units, incorrect rounding
R Rounding off
J Justification/Reason/Explain
NP No penalty for rounding OR omitting units

This marking guidelines consists of 6 pages . NATIONAL SENIOR CERTIFICATE
MATHEMATICAL LITERACY P2

MARKING GUIDELINES
JUNE 2018

GRADE 11 NATIONAL SENIOR CERTIFICATE

Demo
NW/JUNE/MATLIT/ EMIS/6*******

Ques Solution Explanation TL

QUESTION 1 [18 MARKS]

1.1.1 P = R450 + R0,50 ×250■
= R450 + R125
= R575■ 1M m ultiplying and adding
1A answer
AO
(2) F
L2

1.1.2 Q = R200 + R2 ×(200 – 100) ■
= R200 + R200
= R400■ 1M multiplying and adding
1A answer
AO
(2) F
L2

1.2

1A (0;200)

1A any point between
100 km and 300 km

1A (300;600)

1A (350;700)

1A joining the points

1A naming the graphs

1A heading

(7) F

L3

1.3 Breakeven point is the point where the cost of hiring a car for option 1 and option 2 is the same for the same distance travelled. ■■ 2O explanation

(2) F

L4

1.4 Distance = time × average speed

250 km = time × 100 km/h ■

time = 250 ■■

100 ■■/■ ■

time = 2,5 hours ■

2,5 hours = 2 hours 30 minutes ■

Jabulani’s claim is incorrect. ■

1SF substitution

1S change subject of formula

1CA time

1C in hours & minutes

1O opinion

M

L4 0501001502002503003504004505005506006506700
0 50 100 150 200 250 300 350Amount in rand
Distance in kmCost of hiring a car
Option 1
Option 2

Demo
NW/JUNE/MATLIT/ EMIS/6*******

OR
2 hours 45 minutes = 2,75 hours■
Distance = time × average speed
250 km = 2,75h × average speed ■
average speed = 250 ■■
2,75 ■ ■
average speed = 90,9 km/h ■
Jabulani’s claim is incorrect. ■

OR
2 hours 45 minutes = 2,75 hours ■
Distance = time × average speed
Distance = 2,75 h × 100 km/h ■
= 275 km ■■
Jabulani’s claim is incorrect. ■ OR

1C time in hours
1SF substitution
1S change subject of
formula
1CA average speed
1O opinion

OR

1C time in hours
1SF substitution
2CA distance
1O opinion
(5)

QUESTION 2 [24 MARKS]

2.1 Area of the four additional family members= 4 ×0,7 m2
= 2,8 m2 ■
Total area = (2 + 2,8) m2
= 4,8 m2 ■
Area of rectangle = length × breadth
4,8 m2 = length ×1,5 m ■
4,8
1,5 = length■
3,2 m = length ■
1M additional area

1A total answer

1SF substitution

1S changing the subject of
the formula
1CA length
(5) M
L3

2.2 150 ■ = 150 000 cm3■
Volume of cylinder = π × r2 × h
150 000 cm3 = 3,142 × r2 × 120 cm ■

377,04000 150 = r2 ■
397,8357734 = r2 ■
19,94582095 = r
20 cm = r ■ 1C conversion

1SF substitution

1S changing the subject of
the formula
1S simplification
1CA radius
(5) M
L3
2.3.1 Discount = 15
100× R11 5 90
=R1 738 ,50■
Total cost required = R11 5 90 – R1 73 8,50 ■
= R9 851,50■
Her claim is correct. ■

1A discount
1M subtraction
1CA total cost
1O opinion F
L4

OR
100% -15% = 85 % ■
Total cost required = 85
100× R11 5 90■
= R9 85 1,50■
Her claim is correct. ■
OR
1A percentage
1M multiplying with 85%
1CA total cost
1O opinion
(4)
2.3.2
(a) Percentage = 350
10 000×100%■
= 3,5 %■ 1M multiplying and
dividing
1A percentage
(2) F
L2
2.3.2
(b) Total monthly repayments = R764,84 ×36■
= 27 534,24■
Interest = R27 534,24 – R10 000
= R17 534,24 ■ 1MA multiplying the
correct value s
1A total repayment
1CA interest

(3) F
L2
2.3.3 In case of death ■■
OR
Permanent disability ■■
OR
Retrenchment ■■
OR
Diagnosed with critical illness ■■
Accept any relevant reason 2O opinion

(2) F
L4
2.4 ■F = 1,8 × ■C + 32■
140■= 1,8 × ■C + 32■ ■
■C = 108■
1,8■
■C = 60■■ 1SF substitution

1S changing the subject of
the formula
1A temperature
(3) M
L3
QUESTION 3 [17 MARKS]
3.1 Measurement = 5 ,2 cm■ (Accept 5,1 cm to 5,3 cm)
Scale 2 ,1 cm : 300 km ■ (Accept 2 cm to 2, 2 cm)
Actual distance = 5,2 cm ×300 km
2,1 ■■■
= 742, 86 km■
OR

Measurement = 52 mm■ (Accept 51 mm to 53 mm
Scale 2 1 mm : 300 km ■ (Accept 20 mm to 2 2 mm)
Actual distance = 52 mm ×300 km
21 ■■■
= 742,86 km■ 1A measured distance
1A measured scale
1M using scale

1CA actual distance

OR

1A measured distance

1A measured scale
1M using scale
1CA actual distance
NPR (4) MP
L3
3.2 Cost of half full tank = R14,01 ×30■ 1M multiplying
1A cost F
L4

= R420,30 ■
The gauge was properly working ■
OR
No. of litres = ■420 ,30
■14,01 ■
= 30 ■■
The gauge was properly working ■ 1O opinion
OR

1M dividing
1A number of litres
1O opinion
(3)
3.3 Distance covered = 30 ■×100 ■■
9 ■■
= 333 km■

Distance left to East London = 742,86 km – 333 km
= 409,86 km■
Mr Thibedi's claim is not valid. ■ CA from 3.1 & 3.2
1M working with
consumption rate
1CA distance covered

1CA remaining distance
1O opinion (4) MP
L4
3.4 From East London take N2 ■pass Port Elizabeth .■ At Knysna ■
take N12■ to Beaufort W est.

OR

From East London take N2 ■then before you reach Port Elizabeth ■
join N10 .■ Continue straight until you cross N9 then join N1 ■to
Beaufort West .
1A N2
1A Port Elizabeth
1A Knysna
1A N12
OR
1A N2
1A Port Elizabeth
1A N10 and N9
1A N1
(4) MP
L3
3.5 Fixing a tyre burst ■■
OR
Buying food ■■
OR
Going to bathroom ■■
OR
Stretch legs ■■
Accept any relevant reason 2O opinion

(2) MP
L4
QUESTION 4 [16 MARKS]

4.1 Height of the mountain = 3 559 ×12 inches■
= 42 708 inches ■
= 42 708
39,37 metres■
= 1 084,78537 metres
≈ 1 085 metres ■
1C conversion
1A height in inches
1C conversion
1CA height in metres
(4) M
L3
4.2 From 08:30 to 18:00 = 9 hours 30 minutes ■
= 570 minutes■
number of trips = 570 ■■■■■■■
30 ■■■■■■■ ■
= 19 ■
The opetator ’s statement is correct. ■
1A duration
1C conversion
1M dividing with 30
1A number of trips

1O opinion
M
L4

OR
30 minutes = 0,5 hour ■
From 08:30 to 18:00 = 9,5 hours ■
number of trips = 9,5 ■■■■■
0,5 ■■■■ ■
= 19 ■
The controller’s statement is correct. ■ OR
1C conversion
1A duration
1M dividing with 30
1A number of trips
1O opinion
(5)
4.3 8+5 = 13■
Number of adults = 8
13× 65■
= 40 ■ 1A adding ratio values
1M using ratio values
1A number of adults
(3) M
L2
4.4 Total cost = 40 ×7,6 $ + 25 ×4,8 $■
= 424 $■
= R 424
0,08 ■
= R5 300
The cashier’s claim is valid. ■

OR
Adult rate = R 7,6
0,08 = R95 ■
Children’s rate = R 4,8

0,08 = R60 ■

Total cost = R95 × 40 + R60 × 25 ■
= R5 300
The cashier's claim is valid. ■ CA from 4.3
1M multiplying and
adding
1CA cost in $
1C conversion

1O opinion

OR

1A adult rate
1A children's rate

1M multiplying and
adding
1O opinion
(4) F
L4

Demo
NW/JUNE/MATLIT/ EMIS/6*******

MARKS: 75
TIME: 1, 5 hours

This question paper consist of 9 pages including an annexure and an answer sheet

MATHEMATICAL LITERACY PAPER 1
JUNE EXAM 2018
9
m
MEMORANDUM
NATIONAL
SENIOR CERTIFICATE
GRADE 11

Demo
NW/JUNE/MATLIT/ EMIS/6*******

INSTRUCTIONS AND INFORMATION
1. This question paper consists of FOUR questions. Answer ALL the questions.
2. Use ANNEXURE to answer QUESTION 4.2 and the ANSWER SHEET to answer

QUESTION 2.4.

3. Number the answers correctly according to the numbering system used in the question paper.
4. A LL the calculations must be clearly shown.
5. An approved calculator (non - programmable and non - graphical) may be used, unless stated otherwise.
6. Round off ALL final answers appropriately according to the given context, unless stated otherwise.

7. indicate units of measurement , where applicable.
8. Maps and diagrams are NOT necessarily drawn to scale , unless stated otherwise.
9. Write neatly and legibly.

Demo
NW/JUNE/MATLIT/ EMIS/6*******

QUESTION 1

1.1

Mrs Olifant intends to bake muffins using the muffin mixture below.

1.1.1 Convert 500g, the mass of the muffin mixture to kg.
(2)

1.1.2 If one cup of oil 25 0ml and the recipe needs 2
3 cup, how many ml of oil is needed.
Round off your answer to the nearest whole number
(3)

1.1.3 The packet i ndicates that 12 large muffins can be baked using this mixture.
Calculate the total number of muffins that can be baked using two

packets. (2)

1.2 The table below shows the cost of parking in a certain town.

Use the table above to answer the questions that follow .

1.2.1 Write down two time slots during which parking will be for free.
(2)

1.2.2 Calculate the total time spend at this parking lot, if the owner of a car parked
from 8: 45 to 13:25.
(3)

1.2.3 Determine the amount that the owner of the car will pay will pay for parking. (2)

[13]

Number of hours	Cost
0 to 1 hour	FREE
Between 1and 3 hours	R 4,00
Between 3 and 5 hours	R 6,00
Between 5 and 7 hours	R 8,00
Between 7 and 9	R 10,00
From 9 hours	R 12,00
Saturdays before 13:00	R 5,00
Sunday and public holidays	FREE

QUESTION 2

2.1

The youth of Rantsheng township are planning to hold their annual conference in June. They have arranged to use th e civic Centre in the township. A basic fee of R2 400 will be charged for decorations and an additional fee of R50, 00 per person will be charged.

The table below shows the cost of using the civic centre.

TABLE1: Cost of using the civic centre

Number of people	0	10	20	40	50	100
Cost (R)	2400	2900	A	4400	4900	7400

They will also raise funds during this conference by selling raffle tickets. The tickets will be sold at R 130 , 00 each.

The table below shows the income of selling the raffle tickets.

TABLE 2: Income from selling the raffle tickets

Number of people	0	10	20	40	B	100
Cost (R)	0	1300	2600	5200	10400	13000

Use table1and 2 and the information above to answer the questions that follow.

2.1 Write down the formula for calculating the total cost of using the civic c
entre. (2)

2.2 Use the formula to calculate the value of A.
(2)

2.3 Use the formula I = R130, 00 × n to calculate the value of B.
(2)

2.4 Use table 1and 2 to draw two line graphs on the same set of axis on the answer
sheet provided. Clearly label the graph cost. (8)

2 .5 How many tickets must the youth sell in order to breakeven
(2)

[16]

Demo
NW/JUNE/MATLIT/ EMIS/6*******

QUESTION 3

3.1
Katlego has an account with a clothing store.

KATLEGO TSATSI
PO Box 1496
Zeerust
2865 Instalment
R200, 00
Over due

Total due
R200, 00
Date
09/11/2016 Due date
08/12/2016
Amount Balance
Opening balance
R1000, 61
08/11/2016 Interest
R23,73 R1024, 34
08/11/2016 ATM payment R200, 00 R824, 34

09/11/2016 Account protection
plan R20, 56 R844, 90
09/11/2016 Lifestyle magazine R25, 00 R869, 90

Use the statement above to answer the questions that follow.

3.1.1. Write down the opening balance of the account .
(2)
3.1.2 How much did Katlego pay for this month ?
(2)
3.1.3. Which deduction on this account is a not necessity?
(2)
3.1.4. Determine the interest as a percentage of the opening balance .

(2)
3.2 Katlego borrows R13 500, 00 at 12% interest compounded monthly. How much will he pay at the end of 3 months ?
(5)

3.3
3.3.1.
Calculate the cost if Katlego uses 27 kl of water.
(5)
3.3.2. The municipality also charges R44, 82 for refuse removal. The amount was increased by 17%. Calculate the new charge . (2)
3.4 Katlego earns R25 000, 00 per month in the department in which he works. Calculate how much he would earn if he was working in a country which pays in pounds. (Exchange rate R1, 00 = £ 16, 93) (3)
[23]

The municipality in which Katlego lives charges monthly tariffs for water according to the sliding scale shown below. Use the scale to answer questions that follow.

Water Usage	Rate per kilo
0 - 6 kl	Nil
6 – 12 kl	R18, 04
12 – 18 kl	R10, 55
18 – 24 kl	R12, 21
24 – 30 kl	R13, 95
30 – 42 kl	R15, 08
42 – 72 kl	R16, 08
More than 72kl	R17, 28

QUESTION 4

4.1
The grade 11 learners of Sphola secondary are planning to start a vegetable garden . They will use a tank to store water for irrigation . The tank is 2,5m long and its diameter is 150cm.

The following formulae and the conversion may be used:
■■■■■■ = ■■■■
■■■■■■■■ ■■■■ =■■■ (■+■), where ■=■,■■■
1000cm3 = 1litre

4.1.1 Determine the radius of the water tank.
(2)

4.1.2 Calculate the volume of water in litres (to two decimal places) when the tank is full.
(5)

4.1.3 Calculate the surface area (to the nearest ■2).
(4)

4.2 The ANNEXURE shows the map of the Ultra marathon. Use the annexure to answer the questions that follow.

4.2.1 Write the total number of the medical stations and the refreshments stations on the map.
(4)

4.2.2 How many kilometres would an athlete have completed at the 17th refreshment station. (2)

4.2.3 At what point will this marathon end. (2)

4.2.4 The winner of this 56 km marathon completed the race in 3 hours and 15 minutes. Calculate his average speed.
(4

You may use the following formula

Speed = ■■■■■■■■■

■■■■

[23]

ANNEXURE:

Demo
NW/JUNE/MATLIT/ EMIS/6*******
 020004000600080001000012000014000
0 10 20 30 40 50 60 70 80 90 100 110Income and costQUE STION 2.4
Name of learner:________________________________ Class:_______

Demo
NW/JUNE/MATLIT/ EMIS/6*******

MARKS: 75
TIME: 1,5 hour

This question paper consists of 10 pages including 2 annexures and 1 answer sheet.
NATIONAL
SENIOR CERTIFICATE
MATHEMATICAL LITERACY P2

JUNE 2018

GRADE 11 NATIONAL
SENIOR CERTIFICATE

Demo
NW/JUNE/MATLIT/ EMIS/6*******
INSTRUCTIONS AND INFORMATION

1. This question paper consists of FOUR questions. Answer ALL the questions.

2. 2.1 Use the ANNEXURES in the QUESTION PAPER to answer the following questions:

■ ANNEXURE A for QUESTION 2.3
■ ANNEXURE B for QUESTION 3

2.2 Answer QUESTION 1. 2 on the ANSWER SHEET attached.

3. Number the answers correctly according to the numbering system used in the question paper.

4. An approved calculator (non - programmable and non - graphical) may be used, unless stated otherwise.

5. ALL the calculations must be clearly shown.

6. Round off ALL final answers appropriately according to the given context, unless stated otherwise.

7. Units of measurement MUST be indicated, where applicable.

8. Diagrams are NOT necessarily drawn to scale, unless stated otherwise.

9. Write neatly and legibly.

QUESTION 1

Use the information above to answer the questions that follow.

1.1 Calculate the following missing values:

1.1.1 P (2)

1.1.2 Q (2)

1.2 Use the grid provided on the ANSWER SHEET to draw the line graph representing Option 2. Clearly label the two graphs. (7)

1.3 Explain the break-even point in this context. (2)

1.4 Jabulani claims that it took him 2 hours and 45 minutes to travel the distance between Pretoria and Klerksdorp at an average speed of 100 km/h.

Verify, using calculations, whether Jabulani’s claim is correct .

You may use the formula:

Distance = time × average speed (5)
[18]

Jabulani , who lives in Pretoria, wants to hire a car for a trip to Klerksdorp and back. The distance from Pretoria to Klerksdorp is 250 km.

He contacts the company for a quotation and is given two options below.

Option 1: The company charges R450 per day plus 50 cents for every kilometre travelled.

Table 1: Cost of hiring a car using Option 1

Distance (km)	0	50	100	150	200	250	300	350
Cost (R)	450	475	500	525	550	P	600	625

Option 2: The company charges R200 per day for the first 100 km or part thereof and R2 per kilometre for every additional kilometres travelled after the first 100 km.

Table 2: Cost of hiring a car using Option 2

Distance (km)	0	50	100	150	200	250	300	350
Cost (R)	200	200	200	300	Q	500	600	700

Demo

QUESTION 2

Use the information above to answer the questions that follow.

2.1 Mrs Thompson was told that she needs solar panels with an area of 2 m2 for the first two members in the household and thereafter an area of 0 ,7 m2 for each additional member.

Calculate the total length of the solar panels needed by Mrs Thompson. (5)

2.2 The geyser has a vo lume of 150 ■ and a height of 120 cm .
Calculate the radius, to the nearest cm, of the geyser. (5)

Mrs Thompson decided to install a solar geyser on the roof of her house to reduce the electricity bill.

The solar panels use sunlight to heat the water stored in the cylindrical tank. The heated
water can then be used in the house.

There are six people altogether in Mrs Thompson's household.

The solar geyser consists of rectangular solar panels and a cyli ndrical storage tank as shown
in the diagrams below.

You may use the following formulae:
Area of rectangle = length × breadth
Volume of cylinder = π × r2 × h, where π = 3,142
1■ = 1 000 cm3 and π = 3,142 length Solar panels Geyser
breadth = 1,5m

2.3

Use ANNEXURE A and the information above to answer questions that follow.

2.3.1 Mrs Thompson claims that a loan of R10 000 will be sufficient to
pay the total cost required by the electrician.

Verify, using calculations, whether her claim is correct. (4)

2.3.2 Determine:

(a) the initiation fee as a percentage of the loan amount. (2)

(b) the total interest she will pay for her loan. (3)

2.3.3 Give ONE possible reason for paying monthly balance protection insurance fee. (2)

2.4 The electrician told Mrs Thompson that the temperature of the geyser should always be at 140 ■F.

Calculate the temperature of the geyser in ■C.

You may use the formula:

■F = 1,8 × ■C + 32■ (3)

[24]

The electrician charges Mrs Thompson R11 590 to supply and install the geyser. Mrs Thompson is offered 15% discount if she pays full amount before installation. She then considered taking a personal loan over 36 months in order to install the geyser. She obtained the personal loan repayment table from Leballo Financial Assistance provided on ANNEXURE A

QUESTION 3

Use ANNEXURE B to answer the questions that follow.

3.1 Calculate the actual distance (in kilometres) between Durban and East London. (4)

3.2 Before refuelling , the fuel gauge indicated that the tank was half full.

Verify, showing calculations, whether the fuel gauge was properly
working. (3)

3.3 Mr Thibedi claims that by the time they refuelled the tank, they were
left with 300 km to East London.

Verify, using the answer in QUESTION 3.2 , whether Mr Thibedi’s claim is valid. (4)

3.4 Describe, in detail, the shortest possible route using the national roads to
travel from East London to Beaufort West. (4)

3.5 Give ONE other reason why the family had to stop at the petrol station. (2)
[17]

Mr Thibedi and his family travelled from Durban to East London. They left Durban
with a full tank of petrol. Along the way they stopped at a petrol station to refuel
at the
cost of R420,30.

Note:
The car has a consumption rate of 9 litres per 100 km.
The capacity of the tank is 60 litres.
The cost of fuel is R14,01 per litre .

A map of South Africa showing the national roads is given on ANNEXURE B

QUESTION 4

Use the information above to answer questions that follows.

4.1 Calculate the height (to the nearest metre) of the mountain. (4)

4.2 The operator states that each cable car makes more than 15 trips in a day .

Verify, using calculations, whether the operator ’s statement is correct. (5)

4.3 If the ratio of adults to children per ride is 8:5, calculate number of adults per cable car carrying the maximum capacity. (3)

4.4

The cashier claims that each cable car carrying a maximum capacity gives a company R5 300 per trip.

Verify, using calculations, whether the cashier's claim is valid. (4)
[16]

TOTAL MARK: 7 5

The cable car travels up and down the mountain which is 3 559 feet above sea level every
15 minutes, ferrying up to a maximum of 65 visitors at a time. The cable car stays open from 08:30 to 18:00. It takes each cable car 30 minutes per trip.

Note: 1 feet = 12 inches
1 metre = 39,37 inches

The rates for riding in a cable car are given as follows:

Adults: 7,6 $
Children below 14 years: 4,8 $

Note: R1 = 0,080 $

ANNEXURE A

QUESTION 2.3

LEBALLO FINANCIAL ASSISTANCE

Personal loan repayment table

Loan amount	MONTHLY PAYMENT FOR DIFFERENT PERIODS				
	12 months	24 months	36 months	48 months	60 months
R4 000	R936,43	R519,28	R384,42	R315,60	R276,76
R10 000	R1 872,85	R1 038,55	R764,84	R613,21	R553,52
R20 000	R2 809, 28	R1 557,83	R1 147,27	R996,81	R830,27
R30 000	R3 745,70	R2 077,10	R1 529,69	R1 262,42	R1 107,03

The amounts shown above are approximates and will vary according to interest rate fluctuations.

Note:

- Interest rate used to calculate amounts provided in the above table is 21% per annum
- The monthly payment excludes the monthly service fee of R75,00 and a monthly balance protection insurance fee of R20,50.
- A once -off initiation fee of R350,00 is payable for new loans.

Demo

ANNEXURE B

QUESTION 3

A MAP OF SOUTH AFRICA SHOWING THE NATIONAL ROADS

0 300 km
10

Demo
NW/JUNE/MATLIT/ EMIS/6*******

ANSWER SHEET

QUESTION 1.2

NAME OF LEARNER:________________________________ CLASS:________

050100150200250300350400450500550600650700
0 50 100 150 200 250 300 350Amount in rand
Distance in km_____________________________________